

# Watermark Forgery Attack

atml\_team072

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## 1 Introduction

The goal of this task is to forge invisible image watermarks. For each of eight unknown watermarking schemes we are given 25 genuine watermarked images that all carry the same hidden message, and we must stamp that watermark onto 200 clean target images. Each forged image is scored as  $S_{\text{final}} = S_{\text{det}} \cdot S_{\text{qit}}$ , where  $S_{\text{det}} = \max(0, 2(\text{BitAcc} - 0.5))$  rewards a detector recovering the correct message and  $S_{\text{qit}} = e^{-8 \cdot \text{LPIPS}}$  penalises perceptual distortion. Success therefore requires both faithfully reproducing a scheme’s watermark and keeping the forged image visually close to the clean target. We operate under near black-box conditions: we never observe the schemes or their detectors, only 25 example outputs per scheme.

## 2 Approach

Our central idea is that the eight groups belong to three families, each needing a different attack. We first diagnose each group’s family from the statistics of its 25 sources, then apply the matching method.

**Diagnosis.** For every group we measured the *residual coherence* of the 25 sources—the fraction of high-pass residual energy shared across images—sweeping all frequency bands. Groups WM\_4, WM\_5, WM\_6 showed a strong shared additive component (coherence 1.5–2.1), whereas WM\_1, WM\_2, WM\_3, WM\_7, WM\_8 sat at the noise floor ( $\approx 1.0$ ). Coherence alone is not decisive (the learned codec RivaGAN also shows none), so for the low-coherence groups we ran candidate public decoders and tested whether the recovered message is consistent across all 25 sources relative to a clean-image baseline. Image resolutions gave a further cue, matching published defaults (HiDDeN 128<sup>2</sup>, RivaGAN 256<sup>2</sup>, Stable Signature 512<sup>2</sup>).

**Averaging (copy) attack: WM\_4, WM\_5, WM\_6.** For the additive-coherent groups we extracted the shared watermark as the per-pixel median of high-pass residuals over the 25 sources, choosing the high-pass band per group by maximising matched-filter separation ( $d'$ ). We added this template to each clean target with strength set by

projection-matching to the genuine sources, times a global factor `GLOBAL_SCALE = 2.0` tuned on the leaderboard. This realises the averaging attack of Yang et al. yang2024: since the message is constant, content cancels across sources while the watermark survives. `WM_5` ( $128^2$ ) is consistent with `HiDDeN`; `WM_4` and `WM_6` are additive learned marks forged by the same procedure.

**Decoder identification and native re-embedding: `WM_1`, `WM_2`, `WM_7`.** For schemes with public detectors we recover the message and re-embed it with the scheme’s own encoder, giving high bit accuracy at low distortion. A scheme is identified by the property that its correct decoder returns the same bits from all 25 sources. **`WM_1 = DwtDct`:** the decoder read a consistent message, but consistency aliased across several candidate lengths; we cracked the true length (48 bits) by *signature matching*—only when a mark synthesised at the true length reproduces the sources’ full consistency-vs-length curve is the length correct. **`WM_2 = RivaGAN`** (32-bit) and **`WM_7 = TrustMark`** (variant Q) were identified by cross-source message consistency and re-embedded with their encoders. Each re-embed was verified locally (forged images decode to the sources’ message) before submission.

**No-box diffusion forgery: `WM_3`, `WM_8`.** These resisted both the copy attack (no coherent residual in any band) and every public decoder we tried (`DwtDctSvd`, `RivaGAN`, `TrustMark`, `WAM`, `VINE`, and the Stable Signature 48-bit extractor), each returning noise near the clean baseline. We therefore applied `WMCopier` dong2025, a no-box attack requiring no decoder: an unconditional DDIM diffusion model is trained on the 25 watermarked sources, and the watermark is injected into a clean target by shallow DDIM inversion (to step 20 of 100) followed by refined denoising. To control distortion we blend the forged output with the clean image,  $x_{\text{forged}} = x_{\text{clean}} + \alpha (x_{\text{denoised}} - x_{\text{clean}})$ , and tune  $\alpha$ . With  $\alpha = 0.35$  the `WM_3` forgery raised the public score, confirming the attack succeeds even from only 25 samples. The main limitation is data: `WMCopier` is designed for thousands of collected images, so with 25 the diffusion model is a weak reconstructor and  $\alpha$  trades detection strength against image quality. `WM_8` was attacked with the same pipeline (trained at  $256^2$  and upscaled to its native  $512^2$ ); the upscaling weakened the injected mark below the detection threshold, so `WM_8` was not successfully forged within our compute budget.

### 3 Results

| Submission (cumulative)                                                        | Public score |
|--------------------------------------------------------------------------------|--------------|
| Copy attack only ( <code>WM_4/5/6</code> )                                     | 0.334        |
| + <code>WM_1</code> native ( <code>DwtDct</code> , length 48)                  | 0.368        |
| + <code>WM_7</code> native ( <code>TrustMark Q</code> )                        | 0.494        |
| + <code>WM_2</code> native ( <code>RivaGAN</code> )                            | 0.624        |
| + <code>WM_3</code> no-box forgery ( <code>WMCopier</code> , $\alpha = 0.35$ ) | <b>0.632</b> |

Native re-embedding gave the largest gains, achieving high bit accuracy at very low LPIPS. The copy attack’s contribution is bounded by its three additive groups, and a GLOBAL\_SCALE sweep confirmed 2.0 to be near-optimal. The WMCopier attack on WM\_3 demonstrably added detectable signal from only 25 samples; its per-group gain is smaller because the low-data diffusion model forces a conservative blend.

## 4 Conclusion

Watermark forgery has direct real-world consequences. If an attacker can lift a provider’s watermark from genuine content and stamp it onto arbitrary images, the watermark no longer proves provenance: fabricated or harmful content can be laundered to appear authentic, and a provider can be framed as the source of material it never produced. Our results show this is easy for post-hoc additive and classical codecs— a simple averaging attack, or recovering and re-embedding with a public decoder, suffices—and that even schemes with no accessible decoder are threatened by no-box, distribution-based attacks such as WMCopier. Robust deployments should keep detectors secret, bind the mark to the content, and avoid relying on fixed public extractors; on its own, watermark detection is not a trustworthy signal of origin.

## Code Repository

<<https://github.com/shreyakolhapure9130-byte/Watermark-Forgery-Attack.git>>

## References

- [1] P. Yang, H. Ci, Y. Song, M. Z. Shou. Can Simple Averaging Defeat Modern Watermarks? NeurIPS 2024.
- [2] Z. Dong et al. WMCopier: Forging Invisible Image Watermarks on Arbitrary Images. NeurIPS 2025.
- [3] T. Souček et al. Transferable Black-Box One-Shot Forging of Watermarks via Image Preference Models. NeurIPS 2025.
- [4] S. Craver, N. Memon, B.-L. Yeo, M. M. Yeung. Resolving rightful ownerships with invisible watermarking techniques. IEEE JSAC 1998.
- [5] M. Kutter, S. Voloshynovskiy, A. Herrigel. The Watermark Copy Attack. SPIE 2000.